



## CSA0501-DATABASE MANAGEMENT SYSTEMS

### LAB EXPERIMENTS

DATE: 20/07/2026

### EXPERIMENT 6 -- SELECT with various clause – WHERE, pattern matching

#### AIM:

To retrieve records from the **student\_cgpa** table using the **SELECT** statement with **BETWEEN**, **IN**, and **Aggregate Functions** in MySQL Workbench.

#### PROCEDURE:

1. Open MySQL Workbench and connect to the MySQL server.
2. Select the required database using the USE command.
3. Create the student\_cgpa table and insert the required records.
4. Verify the table contents using the SELECT statement.
5. Execute queries using the **BETWEEN** operator to retrieve records within a specified range.
6. Execute queries using the **IN** operator to retrieve records matching specific values.
7. Use aggregate functions such as **COUNT()**, **SUM()**, **AVG()**, **MIN()**, and **MAX()** to perform calculations on the table data.
8. Observe the output of each query and verify the results.
9. Conclude the experiment after successful execution.

#### PROGRAM:

##### 1. CREATE TABLE

```
CREATE TABLE student_cgpa (  
  reg_no VARCHAR(15) PRIMARY KEY,  
  student_name VARCHAR(50) NOT NULL,  
  department VARCHAR(20),  
  no_of_subs_pass INT,  
  no_of_subs_fail INT,  
  cgpa DECIMAL(3,2)  
);
```

##### Then Insert Values:

```
INSERT INTO student_cgpa  
(reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa)  
VALUES  
(192411123, 'Deepthi', 'CSE', 22, 2, 8.71),  
(192512121, 'Varun', 'IT', 27, 1, 9.23),  
(192312117, 'Aadhya', 'ECE', 25, 3, 8.97),  
(192411227, 'Dimple', 'CSE', 31, 0, 9.37),
```

```
(192820896,'Mohan','EEE',18,9,6.48),
(192411122,'Yogitha','CSE',31,1,9.01),
('192516871','Sadhya','IT',25,5,7.22),
(192111101,'Rahul','MECH',30,12,5.14),
(192411010,'Alekhya','CSE',35,2,9.01),
(192511135,'Rahul','CIVIL',37,0,9.72);
```

```
SELECT * FROM student_cgpa;
```

## OUTPUT:

| reg_no    | student_name | department | no_of_subs_pass | no_of_subs_fail | cgpa |
|-----------|--------------|------------|-----------------|-----------------|------|
| 192111101 | Rahul        | MECH       | 30              | 12              | 5.14 |
| 192312117 | Aadhya       | ECE        | 25              | 3               | 8.97 |
| 192411010 | Alekhya      | CSE        | 35              | 2               | 9.01 |
| 192411122 | Yogitha      | CSE        | 31              | 1               | 9.01 |
| 192411123 | Deepthi      | CSE        | 22              | 2               | 8.71 |
| 192411227 | Dimple       | CSE        | 31              | 0               | 9.37 |
| 192511135 | Rahul        | CIVIL      | 37              | 0               | 9.72 |
| 192512121 | Varun        | IT         | 27              | 1               | 9.23 |
| 192516871 | Sadhya       | IT         | 25              | 5               | 7.22 |
| 192820896 | Mohan        | EEE        | 18              | 9               | 6.48 |

| # | Time     | Action                                                                                                    | Message                                                  | Duration / Fetch      |
|---|----------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------|
| 1 | 22:41:47 | CREATE TABLE student_cgpa ( reg_no VARCHAR(15) PRIMARY KEY, student_name VARCHAR(50) NO...                | 0 row(s) affected                                        | 0.031 sec             |
| 2 | 22:44:31 | INSERT INTO student_cgpa (reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa) VA... | 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0 | 0.000 sec             |
| 3 | 22:46:18 | SELECT * FROM student_cgpa LIMIT 0, 1000                                                                  | 10 row(s) returned                                       | 0.000 sec / 0.000 sec |

## BETWEEN Operator

1. Display students whose CGPA is between 8.00 and 9.50.

```
SELECT *
FROM student_cgpa
WHERE cgpa BETWEEN 8.00 AND 9.50;
```

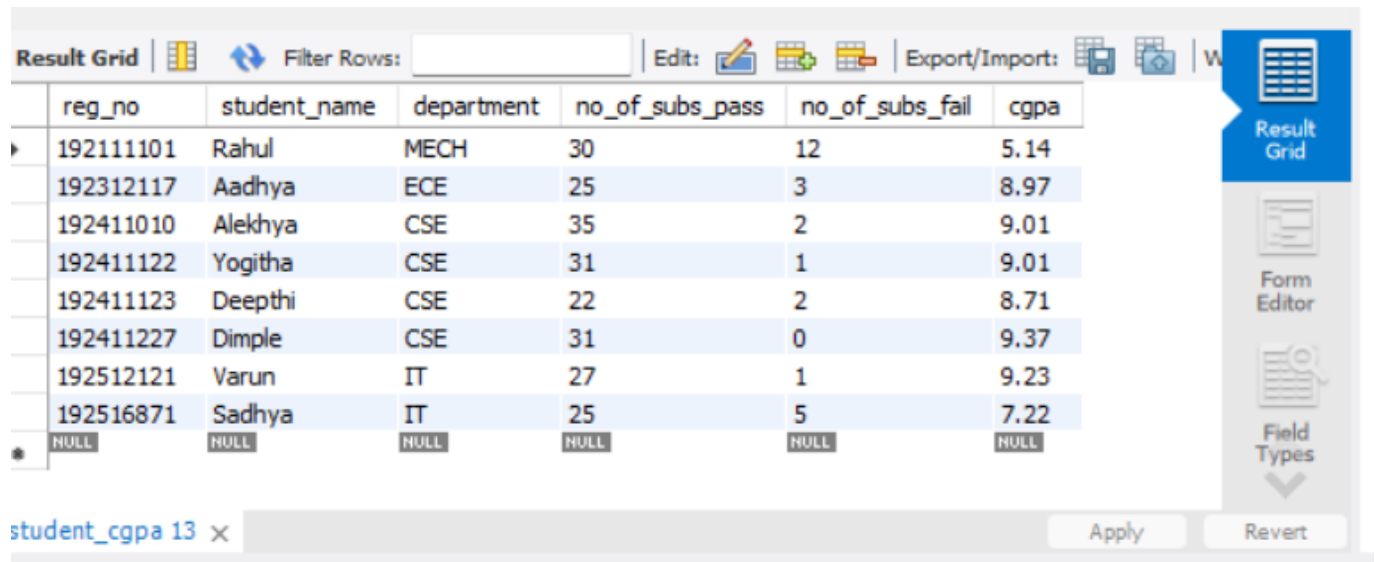
## OUTPUT:

| reg_no    | student_name | department | no_of_subs_pass | no_of_subs_fail | cgpa |
|-----------|--------------|------------|-----------------|-----------------|------|
| 192312117 | Aadhya       | ECE        | 25              | 3               | 8.97 |
| 192411010 | Alekhya      | CSE        | 35              | 2               | 9.01 |
| 192411122 | Yogitha      | CSE        | 31              | 1               | 9.01 |
| 192411123 | Deepthi      | CSE        | 22              | 2               | 8.71 |
| 192411227 | Dimple       | CSE        | 31              | 0               | 9.37 |
| 192512121 | Varun        | IT         | 27              | 1               | 9.23 |
| NULL      | NULL         | NULL       | NULL            | NULL            | NULL |

## 2. Display students who passed between 20 and 35 subjects.

```
SELECT *  
FROM student_cgpa  
WHERE no_of_subs_pass BETWEEN 20 AND 35;
```

### OUTPUT:



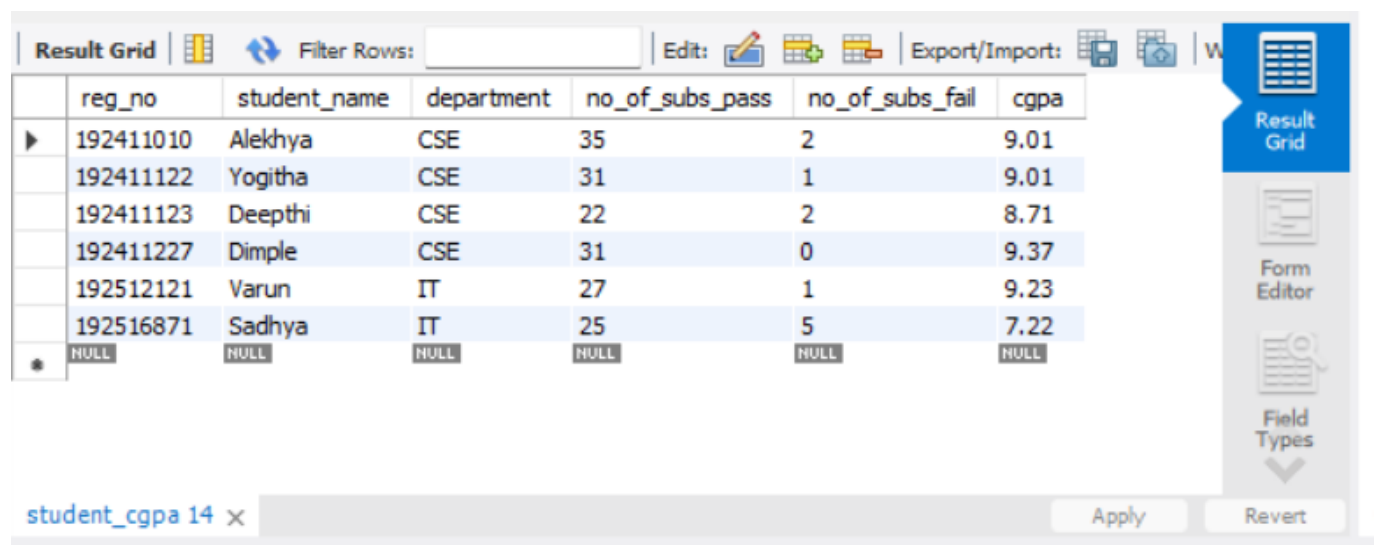
| reg_no    | student_name | department | no_of_subs_pass | no_of_subs_fail | cgpa |
|-----------|--------------|------------|-----------------|-----------------|------|
| 192111101 | Rahul        | MECH       | 30              | 12              | 5.14 |
| 192312117 | Aadhya       | ECE        | 25              | 3               | 8.97 |
| 192411010 | Alekhya      | CSE        | 35              | 2               | 9.01 |
| 192411122 | Yogitha      | CSE        | 31              | 1               | 9.01 |
| 192411123 | Deepthi      | CSE        | 22              | 2               | 8.71 |
| 192411227 | Dimple       | CSE        | 31              | 0               | 9.37 |
| 192512121 | Varun        | IT         | 27              | 1               | 9.23 |
| 192516871 | Sadhya       | IT         | 25              | 5               | 7.22 |
| NULL      | NULL         | NULL       | NULL            | NULL            | NULL |

## IN Operator

## 3. Display students belonging to the CSE and IT departments.

```
SELECT *  
FROM student_cgpa  
WHERE department IN ('CSE','IT');
```

### OUTPUT:

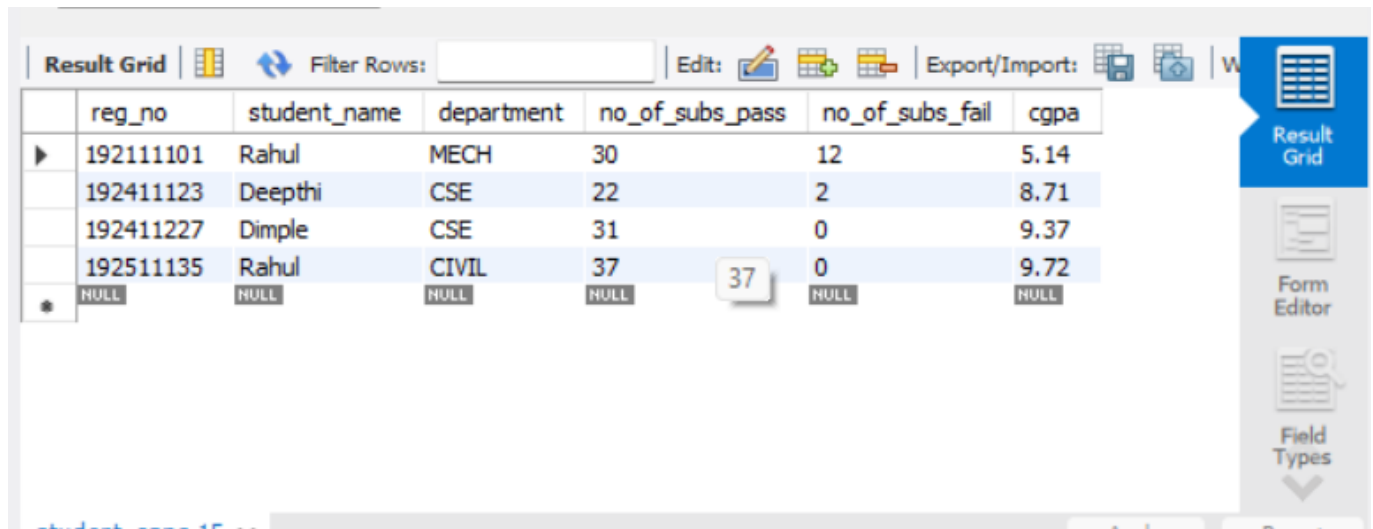


| reg_no    | student_name | department | no_of_subs_pass | no_of_subs_fail | cgpa |
|-----------|--------------|------------|-----------------|-----------------|------|
| 192411010 | Alekhya      | CSE        | 35              | 2               | 9.01 |
| 192411122 | Yogitha      | CSE        | 31              | 1               | 9.01 |
| 192411123 | Deepthi      | CSE        | 22              | 2               | 8.71 |
| 192411227 | Dimple       | CSE        | 31              | 0               | 9.37 |
| 192512121 | Varun        | IT         | 27              | 1               | 9.23 |
| 192516871 | Sadhya       | IT         | 25              | 5               | 7.22 |
| NULL      | NULL         | NULL       | NULL            | NULL            | NULL |

4. Display students whose names are Deepthi, Dimple and Rahul.

```
SELECT *  
FROM student_cgpa  
WHERE student_name IN ('Deepthi','Dimple','Rahul');
```

**OUTPUT:**



The screenshot shows a database application interface with a 'Result Grid' tab. The grid contains the following data:

|   | reg_no    | student_name | department | no_of_subs_pass | no_of_subs_fail | cgpa |
|---|-----------|--------------|------------|-----------------|-----------------|------|
| ▶ | 192111101 | Rahul        | MECH       | 30              | 12              | 5.14 |
|   | 192411123 | Deepthi      | CSE        | 22              | 2               | 8.71 |
|   | 192411227 | Dimple       | CSE        | 31              | 0               | 9.37 |
|   | 192511135 | Rahul        | CIVIL      | 37              | 0               | 9.72 |
| * | NULL      | NULL         | NULL       | NULL            | NULL            | NULL |

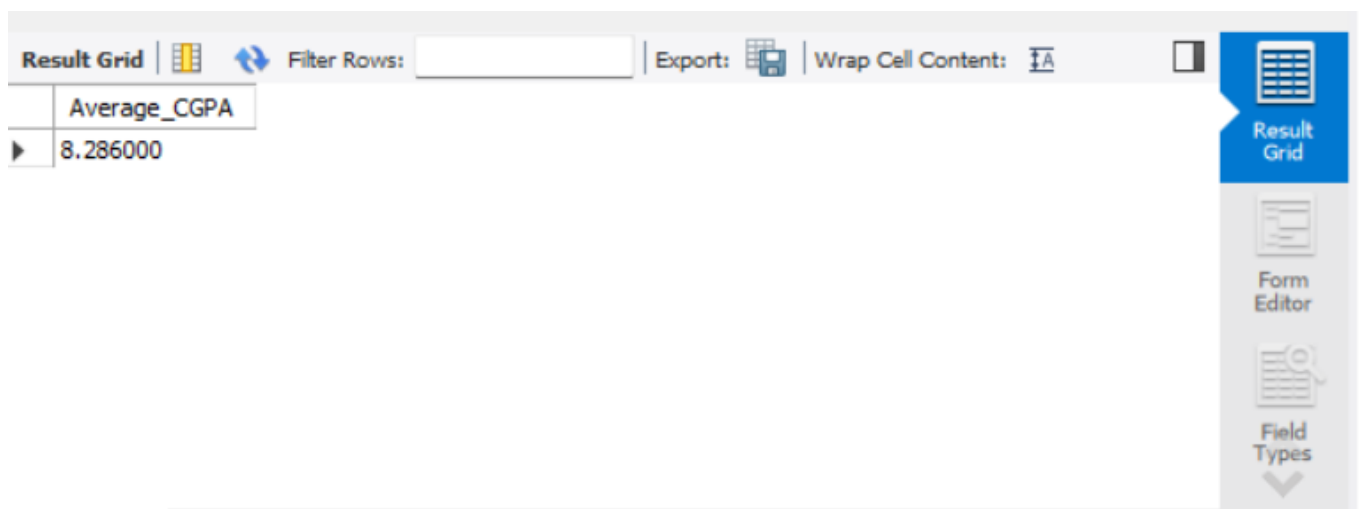
The interface includes a 'Filter Rows' field, 'Edit', 'Export/Import', and 'Wrap Cell Content' buttons. A right-hand sidebar contains 'Result Grid', 'Form Editor', and 'Field Types' options.

## Aggregate Functions

5. Find the average CGPA of all students.

```
SELECT AVG(cgpa) AS Average_CGPA  
FROM student_cgpa;
```

**OUTPUT:**



The screenshot shows the same database application interface, but the 'Result Grid' now displays a single row representing the average CGPA:

|   | Average_CGPA |
|---|--------------|
| ▶ | 8.286000     |

The interface elements are consistent with the previous screenshot, including the 'Filter Rows' field and the right-hand sidebar.

6. Find the maximum and minimum CGPA.

```
SELECT
MAX(cgpa) AS Maximum_CGPA,
MIN(cgpa) AS Minimum_CGPA
FROM student_cgpa;
```

### OUTPUT:

| Result Grid |              |              | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|--------------|--------------|--------------|---------|--------------------|
|             | Maximum_CGPA | Minimum_CGPA |              |         |                    |
| ▶           | 9.72         | 5.14         |              |         |                    |

### 7. Find the maximum, minimum and average CGPA for each department.

```
SELECT
department,
MAX(cgpa) AS Maximum_CGPA,
MIN(cgpa) AS Minimum_CGPA,
AVG(cgpa) AS Average_CGPA
FROM student_cgpa
GROUP BY department;
```

### OUTPUT:

| Result Grid |            |              |              |              | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|------------|--------------|--------------|--------------|--------------|---------|--------------------|
|             | department | Maximum_CGPA | Minimum_CGPA | Average_CGPA |              |         |                    |
|             | MECH       | 5.14         | 5.14         | 5.140000     |              |         |                    |
|             | ECE        | 8.97         | 8.97         | 8.970000     |              |         |                    |
|             | CSE        | 9.37         | 8.71         | 9.025000     |              |         |                    |
|             | CIVIL      | 9.72         | 9.72         | 9.720000     |              |         |                    |
|             | IT         | 9.23         | 7.22         | 8.225000     |              |         |                    |
|             | EEE        | 6.48         | 6.48         | 6.480000     |              |         |                    |

### 8. Display each department and its average CGPA.

```
SELECT
department,
```

```
AVG(cgpa) AS Average_CGPA
FROM student_cgpa
GROUP BY department;
```

### OUTPUT:

| Result Grid |            | Filter Rows: | Export: | Wrap Cell Content: |  |
|-------------|------------|--------------|---------|--------------------|--|
|             | department | Average_CGPA |         |                    |  |
| ▶           | MECH       | 5.140000     |         |                    |  |
|             | ECE        | 8.970000     |         |                    |  |
|             | CSE        | 9.025000     |         |                    |  |
|             | CIVIL      | 9.720000     |         |                    |  |
|             | IT         | 8.225000     |         |                    |  |
|             | EEE        | 6.480000     |         |                    |  |

Result 19 x Read Only

### 9. Calculate the total number of passed subjects.

```
SELECT
SUM(no_of_subs_pass) AS Total_Passed_Subjects
FROM student_cgpa;
```

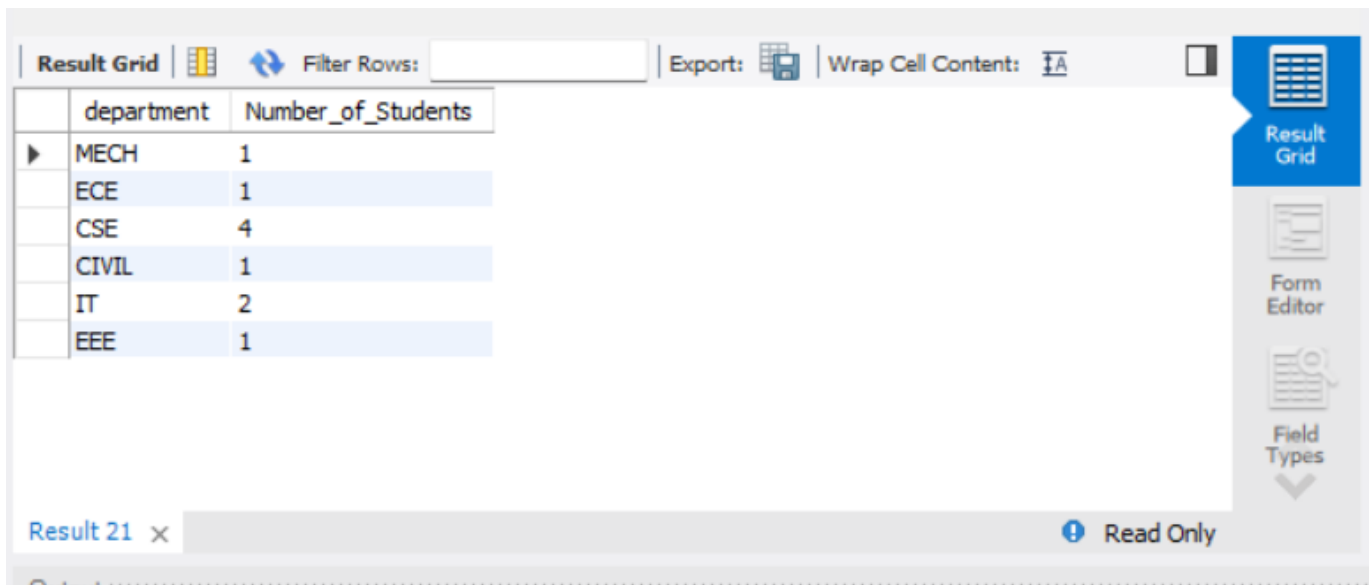
### OUTPUT:

| Result Grid |                       | Filter Rows: | Export: | Wrap Cell |
|-------------|-----------------------|--------------|---------|-----------|
|             | Total_Passed_Subjects |              |         |           |
| ▶           | 281                   |              |         |           |

### 10. Count the number of students in each department.

```
SELECT
department,
COUNT(*) AS Number_of_Students
FROM student_cgpa
GROUP BY department;
```

### OUTPUT:



The screenshot shows a database query result grid with the following data:

|   | department | Number_of_Students |
|---|------------|--------------------|
| ▶ | MECH       | 1                  |
|   | ECE        | 1                  |
|   | CSE        | 4                  |
|   | CIVIL      | 1                  |
|   | IT         | 2                  |
|   | EEE        | 1                  |

The interface includes a toolbar with 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content' options. A sidebar on the right contains 'Result Grid', 'Form Editor', and 'Field Types' buttons. The bottom status bar indicates 'Result 21' and 'Read Only'.

#### 11. Count the total number of students.

```
SELECT COUNT(*) AS Total_Students  
FROM student_cgpa;
```

### OUTPUT:



The screenshot shows a database query result grid with the following data:

|   | Total_Students |
|---|----------------|
| ▶ | 10             |

The interface includes a toolbar with 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content' options.

#### 12. Find the highest CGPA student.

```
SELECT *  
FROM student_cgpa  
WHERE cgpa = (SELECT MAX(cgpa) FROM student_cgpa);
```

### OUTPUT:

Result Grid

Filter Rows:

Edit:

Export/Import:

|   | reg_no    | student_name | department | no_of_subs_pass | no_of_subs_fail | cgpa |
|---|-----------|--------------|------------|-----------------|-----------------|------|
| ▶ | 192511135 | Rahul        | CIVIL      | 37              | 0               | 9.72 |
| ★ | NULL      | NULL         | NULL       | NULL            | NULL            | NULL |

Result Grid

Form Editor

Field Types

### 13. Find the lowest CGPA student.

```
SELECT *
FROM student_cgpa
WHERE cgpa = (SELECT MIN(cgpa) FROM student_cgpa);
```

### OUTPUT:

Result Grid

Filter Rows:

Edit:

Export/Import:

W

|   | reg_no    | student_name | department | no_of_subs_pass | no_of_subs_fail | cgpa |
|---|-----------|--------------|------------|-----------------|-----------------|------|
| ▶ | 192111101 | Rahul        | MECH       | 30              | 12              | 5.14 |
| ✱ | NULL      | NULL         | NULL       | NULL            | NULL            | NULL |

### RESULT:

Thus, the records from the student\_cgpa table were successfully retrieved using the SELECT statement with BETWEEN, IN, and Aggregate Functions in MySQL Workbench.